Downloaded from http://sp.lyellcollection.org/ at University of Iowa on June 9, 2015

Age and duration of the Mississippi Valley-type mineralizing fluid flow event in the Viburnum Trend, southeast Missouri, USA, determined from palaeomagnetism

D. T. A. SYMONS[1], M. T. LEWCHUK[1] & D. L. LEACH[2]

[1]*Department of Earth Sciences, University of Windsor, Ontario, Canada, N9B 3P4*
[2]*United States Geological Survey, MS 973, Box 25046, Federal Center, Denver, CO 80225, USA*

Abstract: The Viburnum Trend is a world-class Mississippi Valley-type (MVT) lead–zinc ore deposit in platform carbonates of the Upper Cambrian Bonneterre Dolomite in the midcontinent of the USA. Palaeomagnetic methods have been used to analyse 233 specimens from early octahedral (nine sites) and late-stage cubic (13 sites) galena ore from four mines along the *c.* 70 km north–south length of the Trend. The characteristic remanence is carried by single to pseudo-single domain pyrrhotite and magnetite. This is the first MVT deposit in which pyrrhotite is shown to be a remanence carrier and present in galena crystals. The remanence directions define an Early Permian mean age of 273 ± 10 Ma for the ore-stage mineralization, a maximum duration for the mineralization event of 12 Ma, and a time difference of 5 Ma between the early octahedral and late cubic galena ore stages. The Early Permian age for the ore is consistent with models of ore genesis that invoke fluid flow from the Ouachita orogen during Ouachitan orogenesis.

Nearly a dozen major Mississippi Valley-type (MVT) lead–zinc-barite deposits have been studied palaeomagnetically over the past few years. They include most of the major past and presently producing districts in North America (Fig. 1) as well as the Silesian deposits in Poland. The majority of this work has been done in the Palaeomagnetic Laboratory at the University of Windsor in Canada, and much of it has been recently summarized (Symons *et al.* 1996). The main conclusions of this work are that: (1) a Phanerozoic MVT mineralization event can be dated to within ± 15 Ma using palaeomagnetism and it is usually a more reliable method than any radiometric technique at present; (2) every MVT event studied to date is coeval with a major orogenic event in the surrounding or adjacent orogen, supporting Oliver's (1986, 1992) assertion that MVT deposits are formed by fluid flow from an orogen; (3) host rock dolomitization may precede the mineralization event by a few million or tens of millions years (Lewchuk & Symons 1995); (4) the epigenetic mineralizing event itself may occur within a few million years of host rock deposition or hundreds of millions of years later; and (5) the duration of the actual mineralization event with associated hydrothermal dolomite lasts from a few million to perhaps 20 million years, in good agreement with hydrogeological models that lead to estimates of 0.5 to 5 Ma to form a major MVT deposit (Sverjensky 1986; Garven *et al.* 1993). The combination of conclusions 1, 3 and 5 above, leads to the possibility that palaeomagnetic methods might be able to absolutely date different stages in the paragenetic or relative age sequence of a large MVT deposit. The Viburnum Trend in southeastern Missouri (Fig. 1) is one of the world's largest known

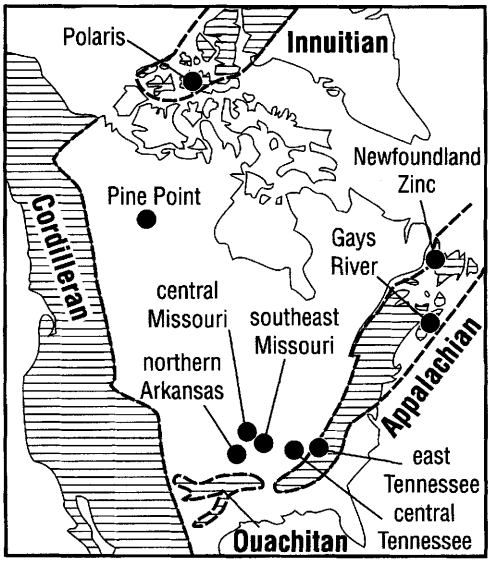


Fig. 1. Locations of major Mississippi Valley-type ore deposits in North America that have been studied palaeomagnetically.

SYMONS, D. T. A., LEWCHUK, M. T. & LEACH, D. L. 1998. Age and duration of the Mississippi Valley-type mineralizing fluid flow event in the Viburnum Trend, southeast Missouri, USA, determined from palaeomagnetism. *In*: PARNELL, J. (ed.) 1998. *Dating and Duration of Fluid Flow and Fluid–Rock Interaction.* Geological Society, London, Special Publications, **144**, 27–39.

Downloaded from http://sp.lyellcollection.org/ at University of Iowa on June 9, 2015

MVT deposits, is being actively mined and thus is readily accessible at many locations, has a paragenetic sequence that is extremely well known (Hagni 1995), has been shown to carry a stable characteristic remanent magnetization (ChRM) (Wisniowiecki *et al.* 1983), and therefore is an obvious target to test the methodology. Within the paragenetic sequence, the start of the main lead–zinc mineralization stage is marked by the formation of octahedral galena crystals with minor cubic modifications and the end of the stage is marked by the formation of cubic galena crystals with rare octahedral modifications. The time interval suggested by this evolution in galena crystallinity is also seen in progressive changes of the fluid inclusion geochemistry (Brannon *et al.* 1991) which are thought to record the geochemical evolution of the fluid flow.

Geology

The southeastern Missouri MVT district has been America's largest producer of lead since 1907 and the world's largest producer since 1970, as well as a significant producer of zinc, copper, silver and cobalt from more than 4.1×10^8 tonnes of ore (Hagni 1995). The Viburnum Trend is one sub-district. This deposit is about 70 km long and up to 300 m wide, and supports ten past and presently producing mines (Fig. 2). From 1960 to 1993 these mines produced about 165×10^6 tonnes of ore with a grade of about 6.0% Pb, 0.6% Zn, 0.2% Cu and 8.6 g/tonne Ag. Hagni (1995) has reviewed the extensive geological literature for the southeastern Missouri MVT district. The following geological notes have been abstracted mostly from his paper, emphasizing only those aspects that are salient to the age of the MVT mineralization.

The Meso-Proterozoic basement rocks are exposed in a circular uplift of about 60 km in diameter in the St Francois Mountains (Fig. 2). The Palaeozoic strata rest unconformably on the basement. Except near minor faults, they dip gently away from the uplift at about 1°. At the base of the sequence are from 0 to 160 m of red sandstones of the Upper Cambrian Lamotte Sandstone. Conformably overlying the sandstones are 80 to 120 m of carbonates of the Bonneterre Dolomite. These Upper Cambrian rocks host the MVT mineralization, setting a maximum age of 514 ± 9 Ma for it. Overlying the Bonneterre Dolomite are a further *c.* 400 m of flat-lying Upper Cambrian to Lower Ordovician dolostones with minor sandstones and shales.

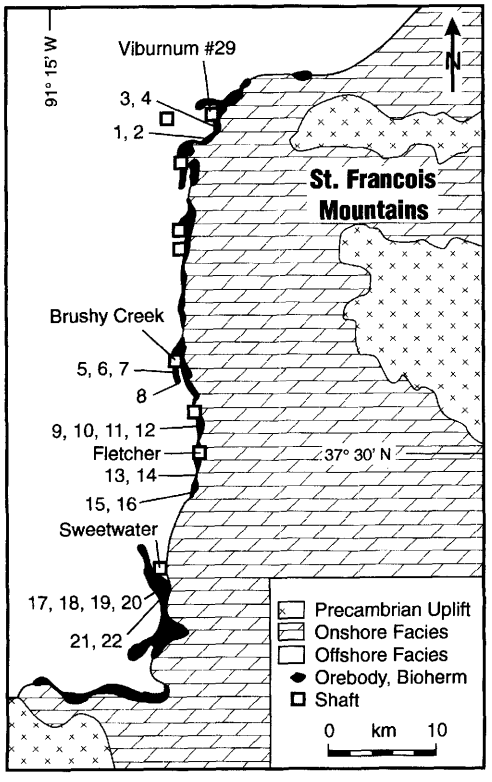


Fig. 2. Locations of mines and sampling sites along the Viburnum Trend in the Bonneterre Dolomite, southeastern Missouri.

The MVT mineralization of the Viburnum Trend occurs in the lower to middle portion of the Bonneterre Dolomite where it follows a fringing algal bioherm and associated oolitic grainstones that were deposited around the St Francois uplift (Gregg *et al.* 1993). Behind the bioherm to the east, the restricted platform or onshore facies is composed of dolomitized burrowed carbonate mudstones interbedded with planar stromatolites. These lithologies are termed 'white rock' locally. The bioherm itself is composed of skeletal and oolitic grainstones with digitate stromatolites. Zones of karst breccia are locally present within the bioherm and immediately above it. In front of the bioherm to the west, the offshore facies is composed of undolomitized grey to brown shaley lime mudstones that are called 'limestones' locally.

The main ore mineral of the Viburnum Trend is galena along with sphalerite and minor chalcopyrite. The gangue is mostly dolomite with marcasite, pyrite, siegenite, calcite and an extensive variety of other minerals as shown in the paragenetic sequence of Hagni (1995). The mineraliza-

Downloaded from http://sp.lyellcollection.org/ at University of Iowa on June 9, 2015

tion is epigenetic and substantially postdates karstification (Gregg *et al.* 1993). It occurs mostly as a 5 to 10 m thick tabular sheet from the massive to disseminated replacement of the host carbonates and as open-space filling in fractures, joints, vugs and karst breccia where it forms the matrix. The earliest generations of galena crystals are termed 'octahedral' but are really cuboctahedral, i.e. octahedrons with varying extents of cubic modification. The early galena crystals are comparatively small, sometimes becoming colloidal, and the associated sphalerite and many other early minerals are also colloidal, thereby indicating rapid precipitation. In contrast the late galena crystals are cubic with rare octahedral modifications. These crystals can be very large with sides exceeding 30 cm in length when found in vugs, indicating slow deposition from dilute solutions. The MVT mineralization in the Trend has been correlated with MVT mineralization found in surrounding districts of the American midcontinent based on regional isotopic, geochemical and paragenetic studies (Leach & Rowan 1986; Leach 1994; Goldhaber *et al.* 1995). In these districts MVT deposits are found in rocks ranging from Upper Cambrian to Mississippian, and traces of MVT mineralization are found in rocks ranging from Precambrian to mid-Pennsylvanian (Coveney 1989). Therefore, unless there were multiple mineralization events throughout the entire time range, the maximum age for MVT ore genesis in the Viburnum Trend is 303 ± 7 Ma.

Radiometric age dating

Heyl *et al.* (1966) attempted to date galena from southeastern Missouri using the Pb–Pb method; however, they found an impossibly old age because the galena contained anomalously radiogenic lead. Using the ^{40}Ar–^{39}Ar method, York *et al.* (1981) obtained an impossibly old age of 549 ± 20 Ma for pyrite and an imprecise age of *c.* 380 Ma for galena. Next Lange *et al.* (1983) reported a Rb–Sr isochron for inclusions in galena of 392 ± 11 Ma; however, the validity of this result was quickly challenged and substantially recanted (Ruiz *et al.* 1985; Lange *et al* 1985). Recently Brannon *et al.* (1991, 1995) showed that the sphalerite has great isotopic heterogeneity, leading them to suggest that the Rb–Sr sphalerite method is an inherently unreliable method for dating the mineralization in the Viburnum Trend.

Indirect dating evidence has been provided by Posey *et al.* (1983), Grant *et al.* (1984) and Stein & Kish (1985). They have reported several Rb–Sr ages for glauconites extracted from the Bonneterre Formation that range between *c.* 350 and 400 Ma. They indicate a maximum age of < 350 Ma for the ore-forming hydrothermal event because it is unlikely that the glauconites have been completely reset. The effect of inheritance is seen more clearly in the results of Hay *et al.* (1995) who reported seven K-Ar illite ages for clays from pods in the MVT ore that range between 297 ± 7 and 489 ± 8 Ma. Their results indicate a maximum age for the ore of < 297 ± 7 Ma.

Palaeomagnetic age dating

Beales *et al.* (1974) and Wu & Beales (1981) were the first to try to date the Viburnum Trend ores using palaeomagnetism. Although they found a few specimens with late Palaeozoic characteristic remanent magnetization (ChRM) directions, they did not have a sufficiently sensitive magnetometer, measure enough specimens, employ sufficiently intense demagnetization methods, nor have a reliable apparent polar wander path (APWP) for comparison for them to determine an accurate age. Subsequently Wisniowiecki *et al.* (1983) did an extensive palaeomagnetic study of the southeastern Missouri district using modern methods. They determined a Pennsylvanian to Early Permian age (289 ± 31 Ma) for the ChRM of the ore but argued on geochemical grounds that magnetite as an oxide would not be coprecipitated with the sulphide ore minerals and, therefore, they concluded that the magnetite with its ChRM was precipitated prior to sulphide mineralization. Noting that Hagni (1995) reported the presence of magnetite in the main ore stage of the paragenetic sequence, Symons *et al.* (1995) argued that the magnetization and mineralization events were coeval, meaning that Wisniowiecki *et al.* (1983) had dated ore genesis.

Sampling and measurement

Three or four oriented block samples were collected at each of 22 sites in the Viburnum no. 29, Brushy Creek, Fletcher and Sweetwater mines (Fig. 2). These mines were chosen to provide a north–south section along the Trend. Sites were selected within these mines to provide excellent examples of octahedral and cubic galena ore in about equal numbers (Table 1). They also include three sites of mineralized breccia clasts from the Sweetwater Mine. A total of

Downloaded from http://sp.lyellcollection.org/ at University of Iowa on June 9, 2015

Table 1. *Site mean remanence directions*

Site, mine	Lithology	Demagnetization Range		Mean ChRM Direction						
		AF (mT)	Thermal (°C)	*b*,	*n*,	n^1	*D*	*I*	α_{95}	*k*
1, V	C	30–90	280–320	4,	11,	10	157	5	8	42
2, V	C	30–130	240–320	7,	9,	8	155	1	9	42
3, V	O	30–90	240–320	4,	11,	11	142	13	8	38
4, V	O	30–110	240–320	4,	11,	11	153	16	8	33
5, B	C	30–90	240–400	4,	11,	11	159	42	15	10
6, B	C	30–90	280–400	4,	11,	11	152	46	20	6
7, B	C	30–90	280–400	4,	11,	11	167	3	6	52
8, B	O	30–110	280–400	4,	11,	11	161	1	5	97
9, F	O	30–130	240–400	4,	11,	10	160	5	9	30
10, F	O	30–90	280–320	4,	11,	11	155	4	8	32
11, F	O	30–130	280–320	4,	11,	10	166	3	10	23
12, F	O	30–130	280–320	4,	11,	10	162	2	9	31
13, F	C	30–90	240–475	4,	11,	11	162	3	4	141
14, F	C	30–130	240–400	4,	11,	11	148	4	5	75
15, F	C	30–130	280–320	4,	11,	11	159	15	10	22
16, F	C	30–130	240–400	4,	11,	11	157	−2	4	122
17, S	C	30–90	240–400	4,	11,	11	145	2	5	105
18, S	Cbx	30–90	240–320	2,	7,	7	169	7	11	34
19, S	Cbx	30–110	240–320	3,	10,	10	161	4	5	104
20, S	Cbx	30–90		3,	9,	8	159	−5	8	47
21, S	O	30–90	280–475	4,	11,	11	159	−4	4	160
22, S	O	30–130	280–400	4,	11,	11	149	5	11	18

Mines: V, Viburnum no. 29; B, Bruahy Creek; F, Fletcher; and, S, Sweetwater.
Lithology: C, cubic galena ore; O, octahedral galena ore; bx, breccia clasts. Mean characteristic remanent magnetization (ChRM) direction: *b*, *n*, n^1, number of blocks, of specimens measured, and of specimens used to calculate the mean. *D*, declination; *I*, inclination; α_{95}, radius of cone of 95% confidence in degrees of arc; *k*, precision parameter (Fisher 1953)

308 specimens of *c.* 2.5 cm right-cylindrical shape were drilled from the 87 oriented blocks. The specimens were stored for about three months in a magnetically shielded room with an ambient magnetic field intensity of *c.* 100 nT to allow their viscous remanent magnetizations (VRM) to decay. All subsequent measurements were done with the specimens remaining inside the shielded room. First, their natural remanent magnetizations (NRM) were measured on an automated Canadian Thin Films DRM-420 cryogenic magnetometer with the capability of resolving a ChRM direction down to an intensity of *c.* 3×10^{-6} A m^{-1}. Next, two to four of the more intensely magnetized specimens were selected for each block to give 11 specimens for nearly all sites and 233 specimens in total for further palaeomagnetic analysis (Table 1).

For each site, two typical specimens were alternating field (AF) demagnetized in 11 steps up to 130 mT using a Sapphire Instruments SI-4 model demagnetizer, and one typical specimen was thermally demagnetized in eight steps up to 525°C using a Magnetic Minerals TD1 demagnetizer. Based on the results from the initial three specimens per site, the remaining specimens in the collection were AF demagnetized at steps of 20, 30, 40, 50, 60, 75 and 90 mT. The uniform demagnetization regimen was deemed desirable because all the specimens showed fairly similar demagnetization characteristics and because it would permit very direct comparison between the octahedral and cubic galena specimens. The ChRM direction for each specimen was determined using the least squares fitting method of Kirschvink (1980) over three or more demagnetization steps and requiring that the maximum angular deviation (MAD) angle be less than 15°. In nearly all specimens the ChRM direction was anchored to the origin.

In addition, 11 specimens representing both ore types and all four mines were subjected to saturation isothermal remanent magnetization (SIRM) testing to better characterize their magnetic mineralogy by progressively pulse magnetizing them in a direct field in seven steps up to 900 mT, and then AF demagnetizing them in six steps up to 130 mT.

As a test to see if the ChRM was being carried in the galena crystals as well as in the dolomite, six artificial specimens were prepared with about the same dimensions as the conventional

Downloaded from http://sp.lyellcollection.org/ at University of Iowa on June 9, 2015

specimens. Two specimens were prepared by mixing a ceramic cement slurry and allowing it to harden in a removable form in a known Earth's magnetic field. These blank specimens were used for control. Two more specimens were prepared by pulverizing several acid-cleaned well-shaped octahedral crystals into a fine powder using a mortar and pestle, and then mixing about 10 g of galena powder into 50 g of cement slurry and forming two more specimens. This last process was repeated using a 'perfect' cube of galena. One blank, one octahedral and one cubic artificial galena specimen were then subjected to the same 11-step AF demagnetization procedure as described above, followed by the same SIRM magnetization and demagnetization procedure. These three artificial specimens were once again pulse magnetized at 900 mT. They were then thermally step demagnetized along with the other three artificial specimens in ten steps up to 600°C. Symons & Sangster (1992) and Symons *et al.* (1992, 1993) used this method previously to show that galena crystals from the Polaris MVT mineralization carry a ChRM in magnetite whereas those from the Pine Point deposit do not carry a ChRM.

Results

Natural remanent magnetization

The median NRM intensities found for the octahedral and cubic galena specimens proved to be very similar at 1.5 and 2.3 $\times$ 10^{-4} A m^{-1}, respectively. These values are typical of NRM intensities found for mineralization in most MVT districts (Symons & Sangster 1994). The NRM directions are mostly scattered along a great circle trend between the present Earth's magnetic field direction, or steeply down to the north, and a south-southeastern and equatorial direction.

Alternating field step demagnetization

Three sorts of AF step demagnetization behaviour were found in both octahedral and cubic galena specimens. The vast majority of specimens showed the removal of a modern VRM that about parallels the present Earth's magnetic field direction up to a cleaning field of *c.* 20 mT, followed by the removal of the ChRM vector in more intense fields (Fig. 3). The rate of intensity

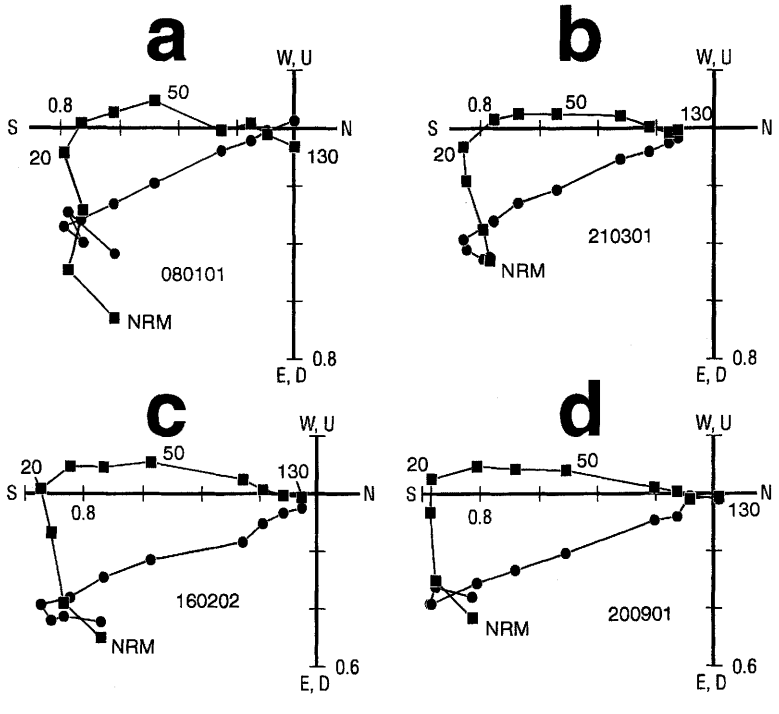


Fig. 3. Vector component plots for the remanence of example specimens of octahedral (**a**, **b**) and cubic (**c**, **d**) galena ore with reversed polarity on alternating field (AF) step demagnetization. Projections onto the horizontal (vertical) north–east–south–west (north-down–south-up) plane are denoted by circles (squares). Axial values are proportionate to the NRM intensity. The AF intensity of some steps are in millitesla (mT).

Downloaded from http://sp.lyellcollection.org/ at University of Iowa on June 9, 2015

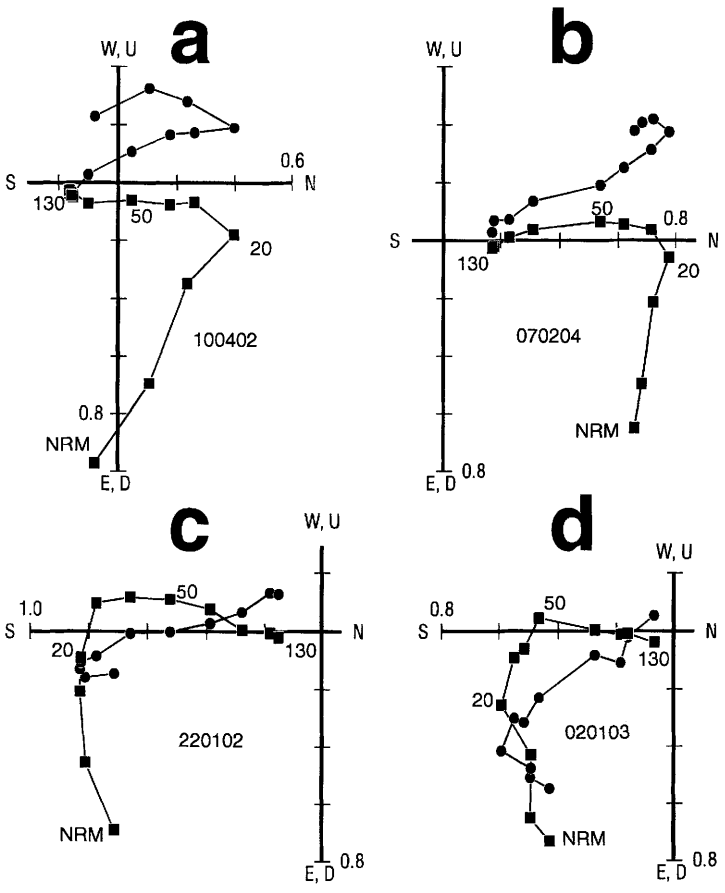


Fig. 4. Vector component plots for example specimens of octahedral (**a, c**) and cubic (**b, d**) galena ore on AF demagnetization with a normal polarity remanence (**a, b**) or a reversed polarity but having an unresolved residual remanence component preventing decay to the origin (**c, d**). Convention as in Fig. 3.

decay is consistent with either magnetite or pyrrhotite being the ChRM carrier. The ChRM direction is usually best isolated in the 30 to 90 mT range, and it is to the south-southeast and nearly horizontal. This ChRM direction defines a reverse polarity of the Earth's magnetic field. Thirteen specimens from three blocks of octahedral ore and one of cubic galena ore follow the same demagnetization pattern except their ChRM direction is about antiparallel to the north–northwest and equatorial or normal in polarity (Fig. 4a, b). The third behaviour was found in all specimens from sites 5 and 6 and occasional other specimens. They also follow the same demagnetization pattern except that they do not decay towards the origin of a component plot, suggesting that an underlying very stable magnetization in goethite or hematite remains after AF demagnetization (Fig. 4c, d). The median intensity of the isolated ChRMs are 4.6 and 9.4 $\times$ 10^{-5} A m^{-1} for octahedral and cubic galena ore, respectively, or about 31% and 41% of their median NRM intensities, respectively.

Thermal step demagnetization

Thermal step demagnetization also showed the initial removal of a modern VRM up to about 240°C, and the isolation of the ChRM at higher temperatures. About half of the specimens showed an initial rapid intensity decrease or unblocking of the ChRM at temperatures between 280 and 320°C that is indicative of pyrrhotite (Fig. 5a, b). This is the first MVT district in which pyrrhotite appears to be one carrier of the ChRM; however, this is not surprising

Downloaded from http://sp.lyellcollection.org/ at University of Iowa on June 9, 2015

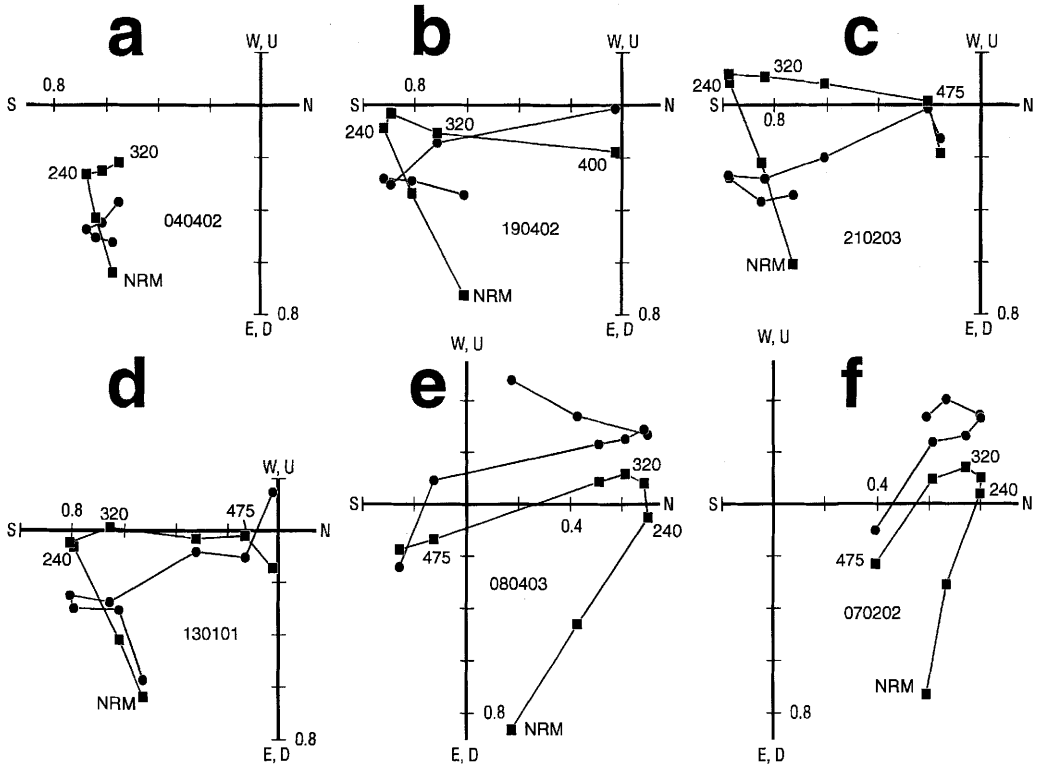


Fig. 5. Vector component plots for example specimens of octahedral (**a**, **c**, **e**) and cubic (**b**, **d**, **f**) galena ore with reversed (**a**, **b**, **c**, **d**) and normal (**e**, **f**) remanence on thermal step demagnetization. The remanence in most pyrrhotite-bearing specimens becomes very (**a**) to moderately (**b**) erratic above 320°C as pyrrhotite oxidizes to form new magnetite. In some specimens the direction is maintained up to 400°C (**c**, **e**, **f**) or even to 475°C (**d**), indicating the presence of Permian magnetite.

because Hagni (1995) has observed it in polished sections as part of the main lead ore stage in the paragenetic sequence. In most mineralized specimens, the sulphides start to oxidize and to grow new magnetite beginning just below 400°C, leading to increasing magnetization intensities with erratic directions. However, most pyrrhotite-bearing specimens retain the same ChRM direction between 320 and 400°C, indicating that ancient magnetite is present, and in several specimens it remains a significant ChRM carrier above 400°C to about 475°C (Fig. 5c, d). Thermal demagnetization also successfully isolates the normal ChRM direction (Fig. 5e, f) in both octahedral and cubic galena ore. In as much as no specimens showed rapid intensity drops between about 80 and 120°C, the unblocking temperature of goethite, it is probable that the very stable remanence remaining after AF demagnetization in specimens from sites 5 and 6 is carried by hematite. The hematite could have a geological origin or have been produced by oxidation on heating in the oven. Unfortunately hematite has a diagnostic unblocking temperature range between 600 and 675°C, and hence its definition is prevented by the generation of new magnetite beginning at about 400°C.

Saturation isothermal remanent magnetization

The SIRM acquisition curves for both octahedral and cubic galena ore specimens show a rapid increase to saturation by about 300 mT (Fig. 6a, c) and AF step demagnetization of the SIRM gives intensity decay curves that are typical of single domain to pseudo-single domain magnetite or pyrrhotite (Fig. 6b, d). Domains in this size range and these minerals usually carry a ChRM that has geological significance.

Downloaded from http://sp.lyellcollection.org/ at University of Iowa on June 9, 2015

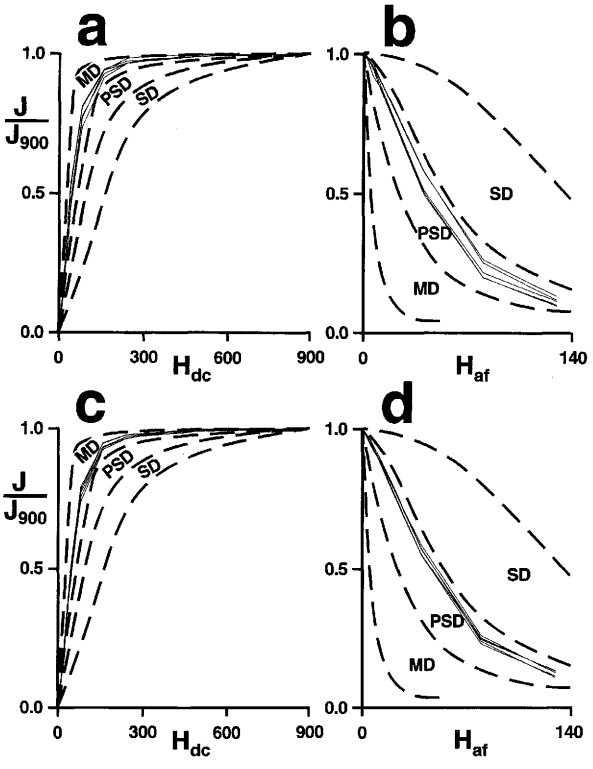


Fig. 6. Saturation isothermal remanent magnetization (SIRM) acquisition (**a**, **c**) and AF demagnetization (**b**, **d**) curves for specimens of octahedral (**a**, **b**) and cubic (**c**, **d**) galena ore. J/J_{900} is the ratio of the measured to SIRM intensity at 900 mT. H is the magnetic field intensity in millitesla. The type curves bound the ranges for single (SD), pseudo-single (PSD) and multidomain (MD) pyrrhotite.

Artificial specimens

The presence of both pyrrhotite and magnetite inclusions in both octahedral and cubic galena crystals is shown clearly in the testing of the artificial specimens. The detrital remanent magnetization created when the powdered galena is allowed to set in a ceramic slurry in the Earth's magnetic field, is preferentially demagnetized in both the pyrrhotite and magnetite diagnostic unblocking temperature ranges (Fig. 7) whereas the blank ceramic control specimens show no such preference. These results are consistent with the direct observation by Hagni (1995) of both pyrrhotite and magnetite in the main ore stage of the MVT paragenetic sequence for the Viburnum Trend, with the thermal demagnetization characteristics of the actual ore specimens, and with the conclusion that the magnetization and MVT mineralization events are coeval for the Viburnum Trend, as Symons & Sangster (1994) concluded.

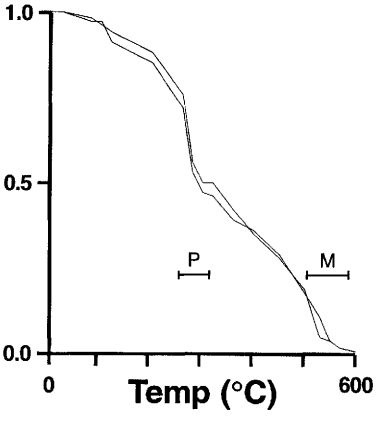


Fig. 7. Thermal demagnetization of artificial ceramic specimens containing powdered octahedral and cubic galena showing the remanence intensity after subtraction of the effect of the ceramic blank. Note the sharp decrease between 260 and 320°C and between 500 and 585°C that are diagnostic of pyrrhotite (P) and magnetite (M) in the galenas, respectively.

Downloaded from http://sp.lyellcollection.org/ at University of Iowa on June 9, 2015

Site mean ChRM directions

When the specimen ChRM directions are grouped by site, they are mostly well-clustered with a radius for their 95% confidence cone (α_{95}) of $\leq 11°$ (Table 1; Fisher 1953). For this calculation, the north-northwesterly ChRM directions were switched in polarity to their antipodal direction. Very few specimens gave aberrant directions, requiring their exclusion from the mean calculation (Table 1). Sites 5 and 6, however, are poorly clustered with α_{95} values of $\geq 15°$ and with ChRM inclinations that are much more steeply inclined than the other sites. Both features suggest that a single ChRM has not been truly isolated, but that two components with overlapping coercivity spectra are present but unresolved. Therefore the results from both sites 5 and 6 have been excluded from further statistical analysis. The remaining sites give a highly clustered grouping of directions (Fig. 8).

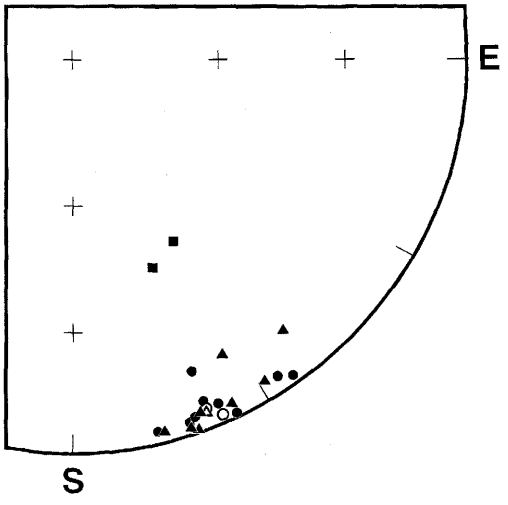


Fig. 8. Southeast quadrant of an equal-area stereogram showing the mean directions for sites of octahedral (triangles) and cubic (circles) galena ore, and of the aberrant magnetizations at sites 5 and 6 (squares). Down (up) vectors are denoted by solid (open) symbols.

Statistical analysis and discussion

Breccia tests

Sites 18, 19 and 20 were collected from karst breccia clasts that showed substantial and varied rotation from their bedding laminations. These clasts carried relatively little galena from replacement; however, they were surrounded by high-grade cubic galena mineralization emplaced by open-space filling after the clasts had fallen into the karst caverns. If the ChRM of the clasts predated mineralization, then their ChRM directions should be randomly oriented. However, this is not the case. The 25 clast mean directions are highly clustered, giving a mean declination (D) of 162°, inclination (I) of 1°, α_{95} of 8°, and precision parameter (k) of 44 (Fisher 1953). This result requires that the magnetization postdates karstification, and implies that the magnetization and mineralization are coeval. This result from the Sweetwater Mine at the south end of the Viburnum Trend also confirms the result of Wisniowiecki *et al.* (1983) from a breccia in the Viburnum no. 28 Mine at the north end.

Pole position and age of the ore

Fisher (1953) statistics assume a circular distribution of vector directions about their mean. Lewchuk & Symons (1995) demonstrated the substantial advantage of using Bingham (1974) statistics, which assume an elliptical distribution of vector directions about their mean, to describe the skewed or streaked distribution given by the site mean ChRM directions of MVT deposits. Following Bingham (1974), the 20 accepted site mean directions give a unit mean direction of $D = 157.3°$, $I = 4.0°$ ($\alpha_{1\text{-}3} = 2.7°$, $\alpha_{1\text{-}2} = 3.9°$, $Az = 111°$) where $\alpha_{1\text{-}3}$ and $\alpha_{1\text{-}2}$ are the semi-minor and semi-major axes of the distribution ellipse and Az is the azimuthal angle of the semi-major axis reckoned clockwise about the mean pole from the longitudinal great circle through the unit mean pole to the south pole. Using the virtual poles for the site mean ChRM directions locates the pole position at 45.2°N, 122.1°E ($\alpha_{1\text{-}3} = 1.4°$, $\alpha_{1\text{-}2} = 3.8°$, $Az = 124°$). When this position and the polar ellipse are plotted on the APWP for North America of Van der Voo (1993) (Fig. 9), they fall within the Early Permian portion of the path. After scaling the path within the Early Permian bounds, this pole provides a direct date of 273 ± 10 Ma for the MVT mineralization in the Viburnum Trend. The path itself has a standard error of about ± 6 Ma during the Early Permian, or significantly less than the calculated error of ± 10 Ma for the age.

Three observations arise from the 273 ± 10 Ma age. First, this result is far more precise but not significantly different from the 289 ± 31 Ma value derived from the result of Wisniowiecki *et al.* (1983). The improved precision is mostly attri-

Downloaded from http://sp.lyellcollection.org/ at University of Iowa on June 9, 2015

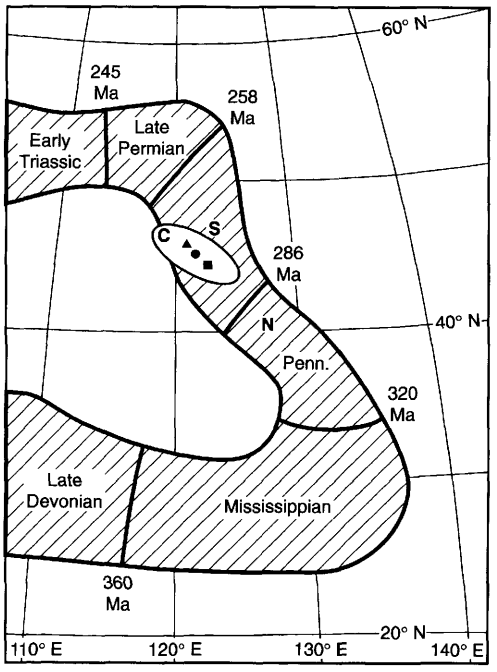


Fig. 9. Apparent polar wander path for the North American craton from Late Devonian to Early Triassic of Van der Voo (1993) showing the mean pole positions for Viburnum Trend ore overall (circle), circumscribed by its Bingham oval of 95% confidence. Also shown are the poles for early-stage octahedral (square) and late-stage cubic (triangle) galena ore, and for the northern (N), central (C) and southern (S) sites along the Trend.

butable to more specific lithological sampling, to measurement of a larger number of specimens, and to use of a more sensitive magnetometer. Second, the age falls within the time span of the Kiaman reversed polarity superchron from *c.* 320 to 250 Ma (Van der Voo 1993), and this is consistent with the finding of a reversed polarity in all but four of our sample blocks. The finding of a normal polarity in three sample blocks of octahedral ore and one of cubic ore, suggests that at least one very short normal polarity event may be present in the Kiaman superchron at *c.* 273 Ma. The third observation is that 273 ± 10 Ma coincides with the Ouachitan orogeny in the adjacent Ouachita orogen, and with the Alleghanian orogeny in the somewhat more distant Appalachian orogen (Fig. 1). The significance of this observation will be discussed later.

Consideration of remagnetization

There is no evidence to suggest that the ChRM in the ore specimens records a thermal remagnetization event between the Late Cambrian and Early Permian. Fluid inclusion homogenization temperatures for the sphalerite and dolomite indicate that the ore was formed between *c.* 150 and *c.* 80°C (Rowan & Leach 1989; Shelton *et al.* 1992). There is no suggestion that the inclusions have been stretched from subsequent reheating to above 150°C. Stratigraphic analysis indicates that burial depths have never exceeded 1.5 km over the Viburnum Trend deposits (Sverjensky 1986) which, using a geothermal gradient of 25°C km^{-1}, gives a maximum burial temperature of *c.* 50°C. Sangster *et al.* (1994) reported an unsuccessful attempt to find conodonts in the Viburnum Trend area; however, the conodont Colour Alteration Index values from the adjacent Upper Mississippi Valley, northern Arkansas and central Missouri MVT districts range between 1 and 2, indicating that the temperature of the host rocks never exceeded 140°C which agrees closely with the inclusion data. This observation is also supported by low values for colour alteration in palynomorphs in the Bonneterre Dolomite (Wood and Stephenson 1989). From the SIRM and thermal demagnetization data, it is evident that the ChRM direction is retained by single to pseudo-single domain magnetite with laboratory unblocking temperatures up to 475°C. Using the curves of Pullaiah *et al.* (1975), recently confirmed by Dunlop (1995) for single domain magnetite, a thermal event of 350°C lasting for 10 Ma would be required to have reset all the ChRMs, and there is no evidence for such an event. Similarly an event of *c.* 250°C for 10 Ma would be required to reset the pyrrhotite ChRMs (D. J. Dunlop, pers. comm., 1997).

Duration of mineralization

Lewchuk & Symons (1995) noted that the elongation direction or skew in the site mean ChRM directions as given by the azimuthal angle of their major axis for an MVT deposit aligned fairly closely to the orientation of the APWP, and that this was caused by apparent polar wander during the period of formation of the deposit. Thus they suggested that the value $V = 2(\alpha_{1\text{-}2} - \alpha_{1\text{-}3})$ provides a measure of the polar wander because $\alpha_{1\text{-}2}$ includes both random variation plus polar wander whereas $\alpha_{1\text{-}3}$ includes only random variation. For the Viburnum Trend, the major axis aligns with the path (Fig. 9) and, therefore, $V = 2(3.8° - 1.4°) = 4.8°$. From the APWP of Van der Voo (1993), the rate of polar wander during the Early Permian is known to be *c.* 2.5 Ma per

Downloaded from http://sp.lyellcollection.org/ at University of Iowa on June 9, 2015

degree of arc. Therefore, the maximum duration for the main mineralization stage from the genesis of octahedral to cubic galena ore is estimated to be 4.8° × 2.5 Ma per degree or 12 Ma using this method.

A second approach to estimating the duration of mineralization is to subdivide the site mean directions into two populations, one of early octahedral and the other of late cubic galena ore sites. They give pole positions of 44.4°N, 122.9°E ($\alpha_{1\text{-}3} = 1.7°$, $\alpha_{1\text{-}2} = 5.9°$, $Az = 143°$) and 45.9°N, 121.4°E ($\alpha_{1\text{-}3} = 1.7°$, $\alpha_{1\text{-}2} = 5.1°$, $Az = 111°$), respectively. Noting that the 95% confidence ellipse about each pole circumscribes the pole of the other population, it is obvious that the two populations are not significantly different at the 95% confidence level. However, the track from the older octahedral to the younger cubic ore does trend in the correct younging direction along the APWP, and the 2.0° angular difference between the two poles gives an estimate for the duration of the mainstage mineralization event of 5 Ma, again using an estimate of *c.* 2.5 Ma per degree for the Early Permian rate of polar wander from Van der Voo (1993).

This result is consistent with: (1) the maximum estimate of ≤12 Ma calculated above; (2) the smallest maximum estimates obtained for other MVT deposits of ≤ 10 Ma for the central Tennessee and central Missouri MVT deposits (Lewchuk & Symons 1995); and (3) the duration of $>2.5 \times 10^6$ years required to form an MVT deposit of Viburnum Trend size that Garven and Freeze (1984) obtained from hydrogeological modelling.

Speculation on fluid flow direction

When the site mean directions for the Viburnum Trend are grouped geographically, they give a pole for the northern sites (sites 1–4) of 40.5°N, 127.4°E ($\alpha_{1\text{-}3} = 3.0°$, $\alpha_{1\text{-}2} = 7.3°$, $Az = 142°$), for the central sites (sites 7–16) of 46.4°N, 119.1°E ($\alpha_{1\text{-}3} = 1.6°$, $\alpha_{1\text{-}2} = 4.0°$, $Az = 115°$), and for the southern sites (sites 17–22) of 46.2°N, 123.3°E ($\alpha_{1\text{-}2} = 2.2°$, $\alpha_{1\text{-}2} = 8.1°$, $Az = 115°$). These poles are not statistically significantly different; however, the pole positions do young from north to south along the Trend (Fig. 9). This trend could be simply random chance, but it could also be an indication that ore genesis proceeded preferentially from north to south along the Viburnum Trend. This result could fit most models of ore genesis such as: (1) the waning of northward flow of mineralizing fluids in the Bonneterre Dolomite from the Arkoma basin because of the Ouachitan orogeny (Leach & Rowan 1986; Gregg & Shelton 1989; Ge & Garven 1992); or (2) southward fluid flow with progressive southerly mineralization of favourable dolomite in the Bonneterre with fluids flowing off the St Francois uplift but coming from the Ouachitan orogen to the south through the Precambrian basement (Goldhaber *et al.* 1995); or (3) northward flow but mineralization focused by deposition in a southward-migrating foreland bulge of the Ouchitan orogeny (Quinlan & Beaumont 1984; Symons & Sangster 1994); or (4) southward flow of one or more fluids out of the Illinois basin (Gregg and Shelton 1989) provided an Early Permian driving mechanism is identified; or (5) mixing of multiple fluids from various sources (Clendenin *et al.* 1994, Plumlee *et al.* 1994).

Conclusions

The following conclusions are drawn from this palaeomagnetic study of the MVT mineralization of the Viburnum Trend.

(1) Thermal step demagnetization and SIRM analyses show that the ChRM of the ore is carried by both magnetite and pyrrhotite. This is the first major MVT district in which pyrrhotite has been found to be a definable ChRM carrier.

(2) In addition to hydrothermal dolomite, both octahedral and cubic galena crystals have been found to carry both a magnetite and a pyrrhotite ChRM based on magnetic testing of artificial specimens carrying powdered galena. This is the first time that pyrrhotite has been found in galena crystals in this fashion and supports the optical observations of Hagni (1995) that both magnetite and pyrrhotite occur in the main ore stage of the paragenetic sequence for southeastern Missouri.

(3) The age of the MVT ore is determined from palaeomagnetism to be 273 ± 10 Ma, indicating Early Permian ore genesis. This improves the previous estimate of 289 ± 31 Ma of Wisniowiecki *et al.* (1983). It is consistent with models that relate ore genesis to fluid flow from the adjacent Ouachita orogen during the coeval Ouachitan orogeny (e.g. Leach & Rowan 1986).

(4) The maximum duration of the MVT mineralization event from the start of precipitation of octahedral galena in the main ore stage of paragenesis to the end of the cubic galena stage is estimated to be 12 Ma using the method of Lewchuk and Symons (1995).

Downloaded from http://sp.lyellcollection.org/ at University of Iowa on June 9, 2015

(5) A second estimate of 5 Ma for the duration of the mineralizing event is given by a comparison of the pole positions for the octahedral and cubic galena ore sties. This estimate agrees closely with those obtained by Garven *et al.* (1993) from hydrogeological modelling.

(6) There is a very speculative hint of a southward younging direction for the MVT mineralization event along the Viburnum Trend, in accord with some models of ore genesis.

The authors thank R. J. Wagner, R. G. Dunn and W. B. Walker for help and the Doe Run Company and ASARCO Incorporated for permission to collect in their mines. We also thank the Natural Sciences and Engineering Research Council of Canada for grant funding to support this research.

References

BEALES, F. W., CARRACEDO, J. C. & STRANGWAY, D. W. 1974. Paleomagnetism and the origin of Mississippi Valley-Type ore deposits. *Canadian Journal of Earth Sciences*, **11**, 211–233.

BINGHAM, C. 1974. An antipodally symmetric distribution on the sphere. *Annals of Statistics*, **2**, 1201–1225.

BRANNON, J. C., COLE, S. C. & PODOSEK, F. A. 1995. Can all Mississippi Valley-Type (MVT) ore deposits be radiometrically dated? Extended Abstracts, International Field Conference of Carbonate Hosted Lead-Zinc Deposits, *Society of Economic Geologists*, 3–7 June 1995, St Louis, Missouri, 27–29.

——, PODOSEK, F. A., VIETS, J. G., GOLDHABER, M. & ROWAN, E. L. 1991. Strontium isotopic constraints on the origin of the ore-forming fluids of the Viburnum Trend, Southeast Missouri. *Geochimica et Cosmochimica Acta*, **55**, 1407–1419.

CLENDENIN, C. W., NIEWENDORP, C. A., DUANE, M. J. & LOWELL, G. R. 1994. The paleohydrology of Southeast Missouri Mississippi Valley-type deposits: Interplay of faults, fluids, and adjoining lithologies. *Economic Geology*, **89**, 322–332.

COVENEY, R. M., JR, GOEBEL, E. & RAGAN, V. M. 1987. Pressures and temperatures from aqueous fluid inclusions in sphalerite from midcontinent country rocks. *Economic Geology*, **82**, 740–751.

DUNLOP, D.J. 1995. Magnetism in rocks. *Journal of Geophysical Research*, **100**, 2161–2174.

FISHER, R. A. 1953. Dispersion on a sphere. *Proceedings of the Royal Astronomical Society of London*, **A217**, 295–305.

GARVEN, G. & FREEZE, A. 1984. Theoretical analysis of the role of groundwater flow in the genesis of stratabound ore deposits. 1. mathematical and numerical model. *American Journal of Science*, **284**, 1085–1124.

——, GE, S., PERSON, M. A. & SVERJENSKY, D. A. 1993. Genesis of stratabound ore deposits in the midcontinent basins of North America. 1. The role of regional groundwater flow. *American Journal of Science*, **293**, 497–568.

GE, S. & GARVEN, G. 1992. Hydromechanical modeling of tectonically driven groundwater flow with application to the Arkoma foreland basin. *Journal of Geophysical Research*, **97**, 9119–9144.

GOLDHABER, M. B., CHURCH, S. E., DOE, B. R., ALEINIKOFF, J. N., BRANNON, J. C., PODOSEK, F. A., MOSIER, E. L., TAYLOR, C. D. & GENT, C. A. 1995. Lead- and sulfur-isotope investigation of Paleozoic sedimentary rocks from the southern midcontinent of the United States: implications for paleohydrology and ore genesis of the southeast Missouri lead belt. *Economic Geology*, **90**, 1875–1910.

GRANT, N. K., LASKOWSKI, T. E. & FOLAND, D. A. 1984. Rb–Sr and K-Ar ages of Paleozoic glauconites from Ohio and Missouri, USA. *Isotope Geoscience*, **2**, 217–239.

GREGG, J. M. & SHELTON, K. L. 1989. Minor and trace element distributions in the Bonneterre Dolomite (Cambrian), southeast Missouri: Evidence for possible multiple basin fluid sources and pathways during lead–zinc mineralization. *Bulletin of the Geological Society of America*, **101**, 221–230.

——, LAUDON, P. R., WOODY, R. E. & SHELTON, K. L. 1993. Porosity evolution of the Cambrian Bonneterre Dolomite, south-eastern Missouri, USA. *Sedimentology*, **40**, 1153–1169.

HAGNI, R. D. 1995. The Southeast Missouri Lead District. *Society of Economic Geologists, Guidebook Series*, **22**, 44–78.

HAY, R. L., LIU, J., BARNSTABLE, D. C., DEINO, A., KYSER, T. K., CHILDERS, G. A. & WALKER, W. T. 1995. Dates and mineralogic results from clay pods of Mine 29 and the Sweetwater Mine, Viburnum Trend, Missouri. Extended Abstracts, *International Field Conference on Carbonate-Hosted Lead-Zinc Deposits*. Society of Economic Geologists, St Louis, Missouri, 3–7 June 1995, 124–126.

HEYL, A. V., DELEVAUX, M. H., ZARTMAN, R. E. & BROCK, M. R. 1966. Isotopic study of galena from the Upper Mississippi Valley, the Illinois-Kentucky, and some Appalachian Valley mineral districts. *Economic Geology*, **61**, 933–961.

KIRSCHVINK, J. L. 1980. The least-squares line and plane and the analysis of palaeomagnetic data. *Geophysical Journal of the Royal Astronomical Society*, **62**, 699–718.

LANGE, S., CHAUDHURI, S. & CLAUER, N. 1983. Strontium isotopic evidence for the origin of barites and sulfides from the Mississippi Valley-type ore deposits in southeast Missouri. *Economic Geology*, **78**, 1255–1261.

LANGE, S., CHAUDHURI, S. & CLAUER, N. 1985. Strontium isotopic evidence for the origin of barites and sulfides from the Mississippi Valley-type ore deposits in southeast Missouri – a reply. *Economic Geology*, **80**, 778–780.

Downloaded from http://sp.lyellcollection.org/ at University of Iowa on June 9, 2015

LEACH, D. L. 1994. Genesis of the Ozark Mississippi Valley-type metallogenic province. *In*: FONTEBOTE, L. & BONI, M. (eds) *Sediment-hosted Zn-Pb Ores*, Berlin, Springer-Verlag, 104–138.

—— & ROWAN, E. L. 1986. Genetic link between Ouachita fold belt tectonism and the Mississippi Valley-type lead–zinc deposits of the Ozarks. *Geology*, **14**, 931–935.

LEWCHUK, M. T. & SYMONS, D. T. A. 1995. Age and duration of Mississippi Valley-type ore mineralizing events. *Geology*, **23**, 233–236.

OLIVER, J. 1986. Fluids expelled tectonically from orogenic belts: Their role in hydrocarbon migration and other geologic phenomena. *Geology*, **14**, 99–102.

—— 1992. The spots and stains of plate tectonics: *Earth Science Reviews*, **32**, 77–106.

PLUMLEE, G., LEACH, D. L., HOFSTRA, A. H., LANDIS, G. P., ROWAN, E. L. & VIETS, J. G. 1994. Chemical reaction path modeling of ore deposition in Mississippi Valley-type deposits of the Ozark region, U.S. midcontinent. *Economic Geology*, **89**, 1361–1383.

POSEY, H. H., STEIN, H. J., FULLAGER, P. D. & KISH, S. A. 1983. Rb–Sr isotopic analysis of upper Cambrian glauconites, southern Missouri: implications for movement of Mississippi Valley-type ore fluids in the Ozark region. *In:* KISVARSANYI, G., GRANT, S. K., PRATT, W. P. & KOENING, J. W. (eds) *Int. Conf. on Mississippi Valley-type lead-zinc deposits. Proc Vol. University of Missouri, Rolla, MO*, 166–173.

PULLAIAH, G. IRVING, E., BUCHAN, K. L. & DUNLOP, D. J. 1975. Magnetization changes caused by burial and uplift. *Earth and Planetary Science Letters*, **28**, 133–143.

QUINLAN, G. M. & BEAUMONT, C. 1984. Appalachian thrusting, lithospheric flexure, and the Paleozoic stratigraphy of the eastern interior of North America. *Canadian Journal of Earth Sciences*, **21**, 973–996.

ROWAN, L. & LEACH, D. L. 1989. Constraints from fluid inclusions on sulfide precipitation mechanisms and ore fluid migration in the Viburnum Trend lead district, Missouri. *Economic Geology*, **84**, 1948–1965.

RUIZ, J., KELLY, W. C. & KAISER, C. J. 1985. Strontium isotopic evidence for the origin of barites and sulfides from the Mississippi Valley-type ore deposits in southeast Missouri – a discussion. *Economic Geology*, **80**, 773–775.

SANGSTER, D. F., NOWLAN, G. S. & MCCRACKEN, A. D. 1994. Thermal comparison of Mississippi Valley-type lead–zinc deposits and their host rocks using fluid inclusion and conodont alteration index data. *Economic Geology*, **89**, 493–514.

SHELTON, K. L., BAUER, R. M. & GREGG, J. M. 1992. Fluid inclusion studies of regionally extensive epigenetic dolomite, Bonneterre Dolomite (Cambrian), southeast Missouri: Evidence of multiple fluids during dolomitization and lead–zinc mineralization. *Bulletin of the Geological Society of America*, **104**, 675–683.

STEIN, H. J. & KISH, S. A. 1985. The timing of ore formation in southeast Missouri: Rb–Sr glauconite dating at the Magmont Mine, Viburnum Trend, southeast Missouri. *Economic Geology*, **80**, 739–753.

SVERJENSKY, D. A. 1986. Genesis of Mississippi Valley-type lead zinc deposits. *Annual Reviews of Earth and Planetary Science*, **14**, 1848–1872.

SYMONS, D. T. A. & SANGSTER, D. F. 1992. Late Devonian paleomagnetic age for the Polaris Mississippi Valley-type Zn-Pb deposit, Canadian Arctic Archipelago. *Canadian Journal of Earth Sciences*, **29**, 15–25.

—— & —— 1994. Paleomagnetic methods for dating the genesis of Mississippi Valley-type lead–zinc deposits. *In*: FONTBOTÉ, L. & BONI, M. (eds) *Sediment-hosted Zn-Pb Ores*. Springer, Berlin, 42–58.

——, PAN, H., SANGSTER, D. F. & JOWETT, E. C. 1993. Paleomagnetism of the Pine Point Zn–Pb deposits. *Canadian Journal of Earth Sciences*, **30**, 1028–1036.

——, SANGSTER, D. F. & LEACH, D. L. 1996. Palaeomagnetic dating of Mississippi Valley-type Pb–Zn–Ba deposits. *In*: SANGSTER, D. F. (ed.) *Carbonate Hosted Lead-Zinc Deposits*. Society of Economic Geologists, Special Publication, **4**, 515–526.

VAN DER VOO, R. 1993. *Palaeomagnetism of the Atlantic, Tethys and Iapetus Oceans*. Cambridge University Press, Cambridge.

WISNIOWIECKI, M. J., VAN DER VOO, R., MCCABE, C. & KELLY, W. C. 1983. A Pennsylvanian palaeomagnetic pole from the mineralized late Cambrian Bonneterre Formation, southeast Missouri. *Journal of Geophysical Research*, **88**, 6540–6548.

WOOD, G. D. & STEPHENSON, J. T., 1989. Cambrian palynomorphs from the warm-water provincial realm, Bonneterre and Davis Formations of Missouri and Arkansas (Reelfoot Rift Area): Biostratigraphy, paleoecology and thermal maturity. *In*: GREGG, J. M., PALMER, J. R. & KURTZ, V. E. (eds) *Field Guide to the Upper Cambrian of southeastern Missouri: Stratigraphy, sedimentology, and economic geology*. University of Missouri-Rolla, Rolla, 84–109.

WU, Y. & BEALES, F. W. 1981. A reconnaissance study by palaeomagnetism of the age of mineralization along the Viburnum Trend, southeast Missouri. *Economic Geology*, **76**, 1879–1894.

YORK, D. A., MASLIWEC, A., HALL, C. M., KUYBIDA, P., KENYON, W. J., SPOONER, E. T. C. & SCOTT, S. D. 1981. *The Direct Dating of Ore Minerals. Ontario Geological Survey, Miscellaneous Paper*, **98**, 334–340.